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SIMULATIONS OF HETEROGENEOUS DETONATIONS AND POST-DETONATION TURBULENT MIXING AND AFTERBURNING

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Abstract. We conduct three-dimensional numerical simulations of the propagation of blast waves resulting from detonation of a nitromethane charge of radius 5.9 cm loaded with aluminum particles and analyze the afterburn process as well as the generation of multiple scales of mixing in the post detonation flow field. In the current study, the particle combustion is observed to be dependent on particle dispersal and mixing of gases in the flow where particle dispersal spreads aluminum within the flow and mixing provides the necessary oxidizer. Thus, 5 μm aluminum particles are burnt more effectively in comparison to 10 μm particles for a fixed initial mass of particles. Also, for a fixed initial particle size, increase in the initial mass of aluminum particles resulted in greater mixing.

Keywords: Detonation, Blast Wave, Afterburn
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INTRODUCTION

Augmentation of impulsive loading is known to occur when solid metal particles are added to an explosive charge, termed heterogeneous explosive, due to the momentum and energy delivered by the particles. When such a charge is detonated, the high-pressure combustion products rapidly expand to drive a blast wave which attenuates due to the effects of spreading [1]. At the same time the metal particles pick up momentum from the gas due to drag, penetrate the contact surface and generate perturbations which grow into hydrodynamic instabilities. These instabilities which are initially Rayleigh-Taylor (RT) [2] in nature are essentially three-dimensional and give rise to mixing of the hot detonation products and the shock-compressed air. Also, as the primary blast wave is driven outward, the flow is over-expanded by the inward moving rarefaction wave, and a secondary-shock is generated. The interaction of the hydrodynamic structures and the secondary-shock results in Richtmyer-Meshkov (RM) instability [3] leading to superior mixing due to vorticity generation by baroclinic effects. Although experimental studies provide evidence of hydrodynamic instabilities in heterogeneous explosions [4], due to the destructive nature of the flow, measurements to characterize the mixing in these explosions may never be possible. Thus, numerical simulations can play a vital role in exploring the details of the mixing and the flow physics, especially in the post-detonation flow.

If the initial charge contains reactive metal particles, the afterburn can provide a significant contribution to the total impulsive loading of the charge [1]. This afterburn is dependent on magnitude of mixing which in turn depends on instability generation. Although previous investigations explored post-detonation flow instabilities due to homogeneous and heterogeneous explosives with inert solid particles [1, 5], the afterburn and mixing in the post-detonation flow of a heterogeneous explosives with reactive solid particles are yet to be explored. Thus, in this paper, we use numerical simulations to study the instability generation and combustion of metal particles ensuing from detonation of a heterogeneous nitromethane charge with aluminum particles. The effects of particle size and the mass loading (η), defined as ratio of mass of particles to mass of the gas

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in the initial charge, on the particle dispersal, mixing and afterburn are also investigated.

NUMERICAL METHOD

Following past numerical studies [1, 5], an Eulerian-Lagrangian formulation is used in the current study to perform three-dimensional simulations of heterogeneous blast waves. The compressible, unsteady, multi-species, multiphase Navier-Stokes equations for reacting gas flow are used to solve for the gas phase whereas Lagrangian tracking approach is used to solve for the position, the mass, the velocity and the temperature of the solid particles. Noble-Abel equation of state is used here for thermodynamic closure [6]. For brevity, the full set of equations, described elsewhere [1], are not presented here. The reaction rates for the problem in the present paper are expected to be very fast, and so, infinite rate chemistry is used to simplify the combustion model following previous studies [1, 7]. The flow is assumed to comprise of 9 species (H_2O, H_2, CO, N_2, O_2, Al, CO_2, Al_2O_3 and AlO) and a five step chemistry employed here is obtained from the six step chemistry presented in [1].

The momentum and energy transfer between the two phases is evaluated using empirical expressions for drag coefficient (C_D) and Nusselt number (Nu) [8]. Following past studies [8, 9], evaporation law specified as, $\frac{dr_p}{dt} = -\frac{r_p}{t_b}\left(1+0.276(Re)^{0.5}\right)$, is used, where r_p denotes the particle radius, Re, the particle Reynolds number, and t_b, the burning time. The burn time data from [9] is used here and the ignition temperature of the aluminum particles is assumed to be 1000 K [8].

The governing equations for the gas phase are solved using MUSCL shock-capturing scheme with HLLC/E Riemann solver [1], and a predictor-corrector scheme for time accuracy. The scheme is second order accurate in both space and time. A 4th order Runge-Kutta scheme is used to solve the solid phase governing equations. Detailed validation of this simulation strategy and numerical approach for shock-turbulence interaction [10], blast waves [1, 5] and detonations [11] has been reported elsewhere.

Three-dimensional sector, with 45° in azimuthal directions, and spherical grids are used in the current study [1] with resolutions $1000 \times 60 \times 60$ and $1000 \times 200 \times 200$, respectively. These resolutions

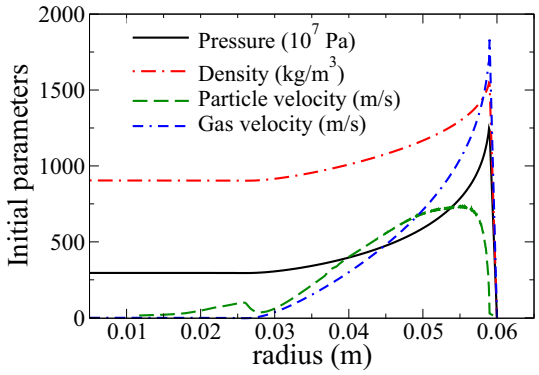


FIGURE 1. Initial detonation profile for heterogeneous nitromethane charge with aluminum particles.

TABLE 1. Initial radius of particles (r_{p_0}), mass loading and number of particles tracked in each case. Particles are tracked by tracking parcels [1], group of particles. Here, each parcel contains 50 particles.

Case	η	r_{p_0} (μm)	Number of parcels
1	1.0	10.0	86881
2	2.0	10.0	173762
3	1.0	5.0	695048

are chosen based on extensive resolution studies performed earlier [1]. The maximum extent in radial direction in both grid configurations is 2.4 m. The initial detonation and particle profiles, shown in Fig. 1, are computed from a 1D simulation based on the Gas-Interpolated-solid Stewart-Prasad-Asay (GISPA) method [12]. Aluminum particles are placed uniformly within the initial charge radius which is 5.9 cm. As indicated in Table 1, η and r_{p_0} are varied and the resulting post-detonation flow is studied in each case.

RESULTS AND DISCUSSION

The detonation of a homogeneous explosive charge results in four critical phases of post-detonation flow which can be summarized as (i) blast wave phase, (ii) implosion phase, (iii) re-shock phase and (iv) asymptotic mixing phase [8]. During the blast wave phase, the initial perturbations at the contact surface of the hot detonation products and the ambient air generate RT instabilities. These

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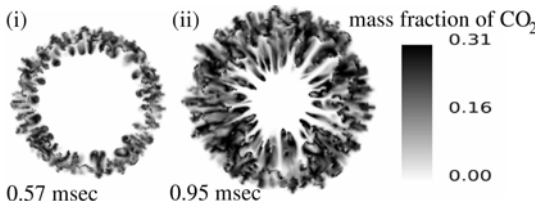


FIGURE 2. Formation of CO_2 due to afterburn in the post-detonation flow of a homogeneous nitromethane charge during (i) blast wave and (ii) implosion phases.

perturbations are considered to occur due to the imperfections in the charge or due to the small scale disturbances arising from molecular scales. In the current study, the perturbations at the contact surface are initialized stochastically based on Gaussian distribution. The mixing of the detonation products with air causes afterburn of CO to generate CO_2. In Fig. 2, the mass fraction of CO_2 is presented to show the mixing layer generated in blast wave and implosion phases. Figure 2 also shows the formation of 'finger-like' projections or spikes due to the expansion of gases towards the center during the implosion phase.

Although simulations performed using spherical grids provide flow features in all directions in great detail, as shown in Fig. 2, these simulations are computationally expensive to perform parametric studies. Thus, we employed sector grids to analyze the effects of η and r_{p_0} on the post-detonation flow. The degree of mixedness obtained using spherical and sector grids is almost identical as shown in Fig. 3 which supports the sector grid configuration employed in the rest of this paper. Here, the degree of mixedness (DM), based on previous studies [8], is defined as $DM = \left[\frac{\int Y_{CO}(Y_{N_2} - Y^i_{N_2})dV}{\int dV}\right] / \left[\left(\frac{\int Y_{CO}dV}{\int dV}\right)\left(\frac{\int (Y_{N_2} - Y^i_{N_2})dV}{\int dV}\right)\right]$, where Y_{CO} and Y_{N_2} are, respectively, the instantaneous mass fractions of CO and N_2 and $Y^i_{N_2}$ is the mass fraction of N_2 in the detonation products.

When a heterogeneous charge explodes the solid particles present within the charge are driven out and as these particles disperse outward they perturb the contact surface between the detonation products and ambient air. Thus, increase in the number of particles results in increased perturbations and hence increase in mixing. Thus, in Cases 2 and 3, due to increased mixing, as a result of more number of particles in

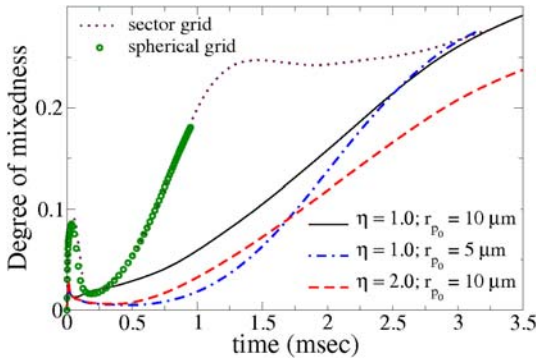


FIGURE 3. Variation of degree of mixedness with time. Dots and circles (sector and spherical grids) indicate cases with homogeneous explosive.

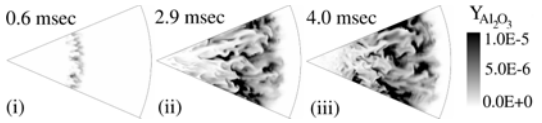


FIGURE 4. Formation of Al_2O_3 due to the combustion of Al in the post-detonation flow of a heterogeneous nitromethane charge (Case 1) during (i) blast wave, (ii) implosion, and (iii) re-shock phases.

comparison to Case1, CO is converted to CO_2 and DM, in Fig. 3, decreases initially till time (t) = 0.5 ms. However, in Case 3, the $5\mu m$ particles burn quickly in comparison to $10\mu m$ particles. The Al combustion consumes O_2 needed for CO_2 production and thus DM in Case 3 increases above the values of DM for Cases 1 and 2 at t = 2.5 ms and 1.75 ms, respectively. In Case 2, the value of DM remains lower than that of Case 1 because of greater mixing and same burn rate for Al. Thus, for a fixed η, DM increased with decrease in r_{p_0} after the implosion phase and for a fixed initial r_{p_0}, DM decreased with increase in η.

As the blast wave is driven outwards, the initial mixing of aluminum particles with air and detonation products results in particle burning and increase in the mass fraction of Al_2O_3 ($Y_{Al_2O_3}$). This is followed by the implosion phase in which the particles are drawn towards the center along with the gases and hence the particle burning is enhanced. As the secondary-shock is formed after the implosion phase, the re-shock drives the particles outwards and the particle burning continues in the mixing layer.

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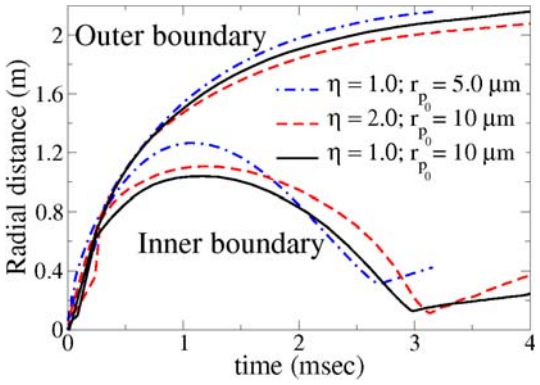


FIGURE 5. Variation of the inner and the outer boundaries of Al combustion with time. Here, the inner most and the outer most radial locations where the mass fraction of Al_2O_3 is atleast 5% of maximum mass fraction of Al_2O_3 is defined as the inner and the outer boundary, respectively.

These phases are shown in Fig. 4. Figure 5 shows the effect of particle dispersion, which determines *Al* availability, and mixing, which determines the amount of oxidizer (O_2 for 'aerobic', and H_2O and CO_2 for 'anerobic' reactions) available, on the particle combustion. In Case 3, even though mixing is greater than Case 1 due to larger number of particles, the burning rate is high and thus the outer boundary overtakes the outer boundary of Case 1. Also, 5 μm particles are entrained and dragged towards center easily in comparison to 10 μm particles. Hence, the inner boundary in Case 3 is drawn below the inner boundary of Cases 1 and 2 during the implosion phase. Again during the re-shock phase, as 5 μm particles have relatively less inertia, the inner boundary overtakes the inner boundaries of Cases 1 and 2. In Case 2, the burn rate of particles is same as in Case 1. Hence higher mixing results in both the inner and outer boundaries of Case 2 to be within the inner and outer boundaries of Case 1. Thus, burn rate and mixing which directly depend on η and r_{p_0} determine the afterburn and particle combustion.

CONCLUSIONS

Three-dimensional numerical simulations of blast wave propagation ensuing from detonation of a heterogeneous nitromethane charge of 5.9 cm radius with aluminum particles are presented. Both homogeneous and heterogeneous explosions generated hydrodynamic instabilities (RT and RM) which resulted in mixing of detonation products and ambient air and enhanced the afterburn. The simulations suggest that the combustion of *Al* is controlled by the particle dispersion and mixing of gases. In each case, mixing controls the amount of oxidizer available for different species to react and the net effect determines the extent of afterburn (*Al* or *CO*).

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